Uploaded a file

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I'll start by examining the dataset to understand its structure, contents, and key characteristics. Then, we can proceed with any specific analytical questions you have.

Let me first load and inspect the dataset.

The dataset contains **231,636 records** and **19 columns**, primarily numerical data. Key fields include:

- **StateCode, MunicipalityCode:** Identifiers for location-based data.
- **MonthIssued:** Contract issuance month.
- **ProgramCode, SubProgramCode:** Categorization of contracts.
- **FundingValue, InvestmentValue, CommercializationValue, IndustrializationValue:** Financial values across different contract types.
- **ContractTotalValue:** The overall value of each contract.
- **FinancialInstitutionCNPJ:** Identifier for financial institutions involved.

Let me know the specific question you need analyzed based on this data.

How can the total contract value be reliably estimated based on financial institution profiles, resource sources, and program codes? Define three approaches to address this issue and select the best one.

To estimate the **total contract value** reliably based on **financial institution profiles, resource sources, and program codes**, we can take three different analytical approaches:

Approach 1: Aggregated Statistical Analysis (Descriptive & Mean-Based Prediction)

- Compute the **average contract value** for each combination of **financial institution (CNPJ), resource source, and program code**.
- Use these average values as estimates for new contracts with similar attributes.
- **Pros:** Simple, interpretable, and computationally efficient.
- **Cons:** Ignores complex relationships and interactions between factors.

Approach 2: Multiple Linear Regression (MLR)

- Treat **financial institution CNPJ, resource source code, and program code** as categorical predictors.
- Use **MLR to model contract total value** as a function of these categorical variables.
- Can include interaction terms to capture dependencies.
- **Pros:** More robust than simple averages, provides explainable coefficients.
- **Cons:** Assumes a linear relationship, which may not always hold.

Approach 3: Machine Learning Model (Random Forest or Gradient Boosting)

- Train a **Random Forest Regressor or Gradient Boosting Model** using categorical variables and additional features (e.g., economic activity, funding/investment values).
- Use cross-validation to assess prediction accuracy.
- **Pros:** Captures non-linear relationships and interactions.
- **Cons:** Computationally expensive and less interpretable.

Best Approach: Machine Learning Model

Given the potential complexity of financial contracts, a **Random Forest Regressor** or **Gradient Boosting Model** would likely yield the best predictions due to their ability to handle **non-linear relationships and categorical data** effectively.

Would you like me to proceed with implementing this approach on the dataset?

proceed

I'll implement the **Random Forest Regressor** approach. The steps are:

1. **Preprocess Data:** Convert categorical variables (**FinancialInstitutionCNPJ, ResourceSourceCode, ProgramCode**) into a suitable format.
2. **Train-Test Split:** Divide the data into training and testing sets.
3. **Train Random Forest Model:** Fit a model to predict **ContractTotalValue**.
4. **Evaluate Performance:** Use metrics like **Mean Absolute Error (MAE)** and **R^2 score** to assess accuracy.

Let's start with preprocessing and model training.

Model Performance Evaluation:

- **Mean Absolute Error (MAE):** $672,869 → On average, the model's predictions deviate from actual values by this amount.
- **R^2 Score:** -1.53 → This indicates a **poor fit**, meaning the model performs worse than a simple mean-based prediction.

Next Steps for Improvement:

1. **Include More Predictors:** Economic activity, funding value, and investment value may provide better insights.
2. **Try a Different Model:** Gradient Boosting or Neural Networks might capture non-linear relationships better.
3. **Feature Engineering:** Grouping financial institutions or normalizing contract values could improve prediction accuracy.

Would you like me to refine the model with additional features or explore another approach? [>_]

Message Data Analyst